Particulate Technology (Powder Metallurgy)

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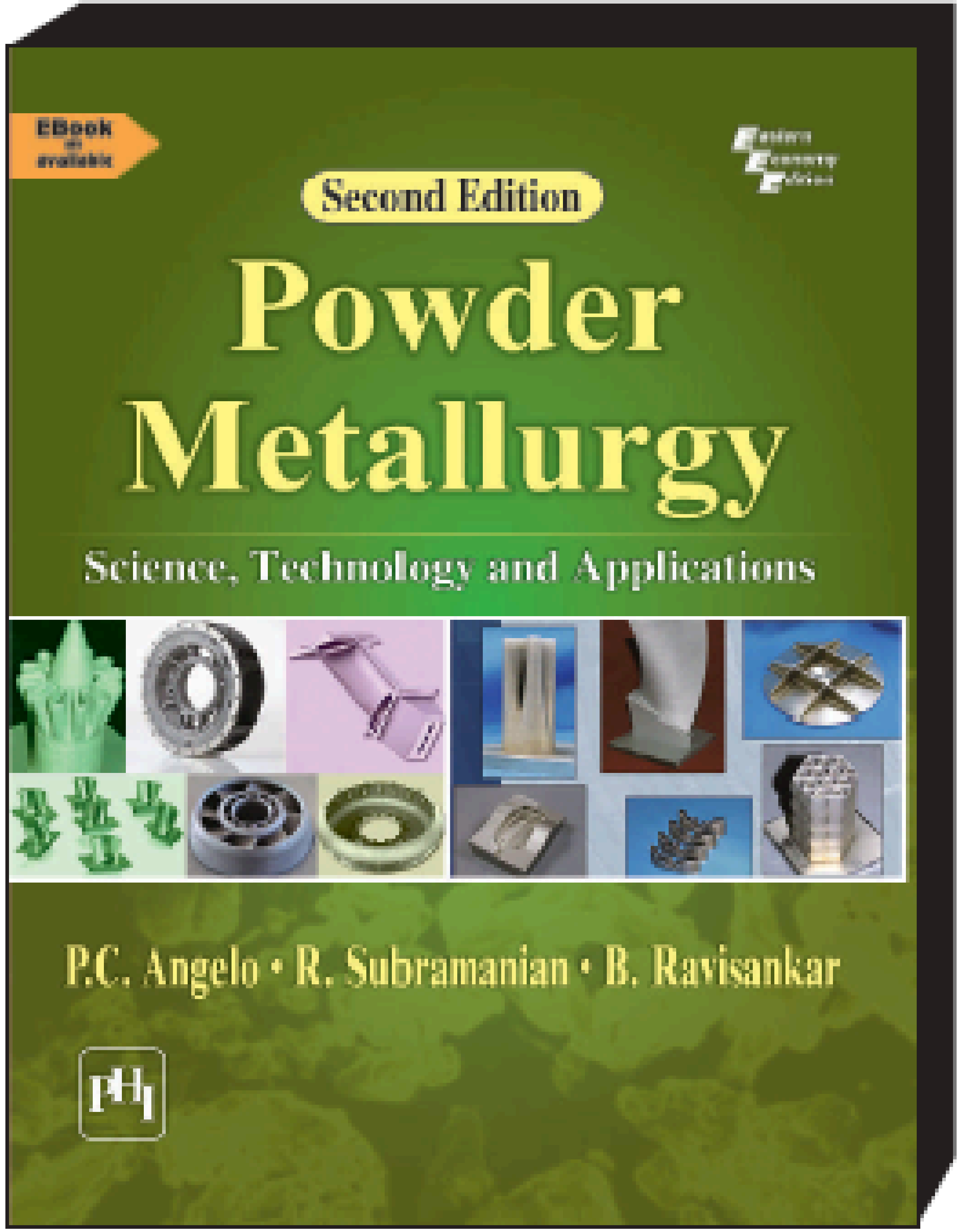
Course file

Mode of Assessment	% Weightage
CT-I	20
CT- II	20
Assignment - I	10
Assignment - II	10
Compensation Assessment	20
End semester Examination	40


EBook available
Second Edition
Powder Metallurgy
Science, Technology and Applications
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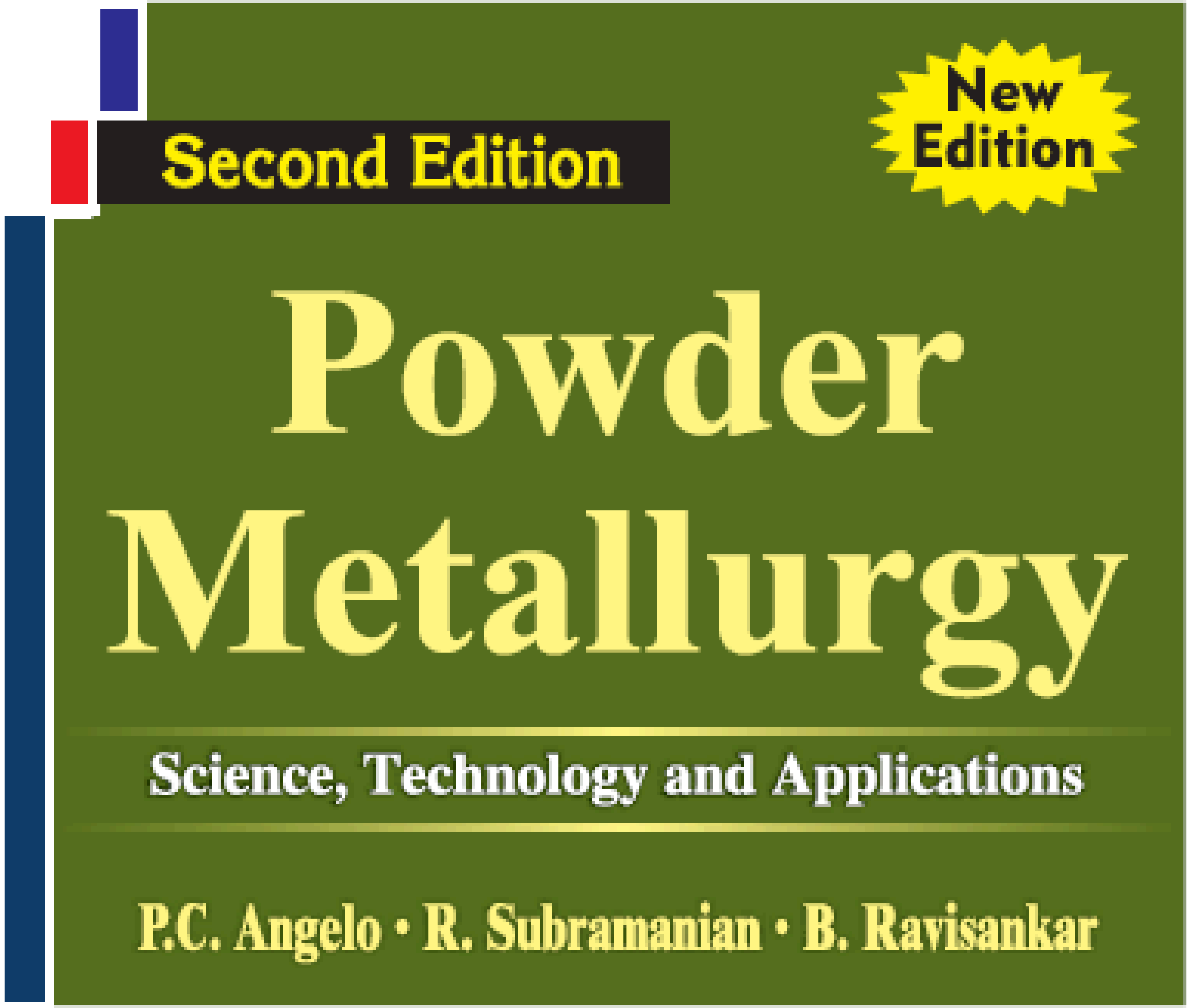

Second Edition
New Edition
Powder Metallurgy
Science, Technology and Applications
P.C. Angelo • R. Subramanian • B. Ravisankar

INTRODUCTION

- ***Powder Metallurgy* (P/M)** has been defined as the art and science of producing metal powders and making semifinished and finished objects from individual, mixed or alloyed powders with or without the addition of non metallic constituents.
- Production of P/M parts involve mixing of elemental or alloy powders with additives and lubricants, compacting the mixture in a suitable die and heating the resulting green compacts in a controlled atmosphere furnace so as to bond the particles metallurgically.
- The resulting parts are solid bodies of material with sufficient strength and density for use in diverse fields of applications.
- Highly-porous parts, precise high-performance components and composite materials can be produced by P/M route.
- P/M offers compositional flexibility, minimized segregation and the ability to produce graded microstructures with varying physical and mechanical properties.
- P/M also offers advantages over ingot metallurgy in terms of quality, cost, precision, productivity as well as by its ability to conserve critical raw materials through a high level of material utilisation.

HISTORICAL BACKGROUND

- Powder Metallurgy, recognized today as one of the most important manufacturing processes for making several industrial products, dates back to a **long time in history.**
- One of the earliest examples of P/M techniques practiced in the ancient days was the reduction of iron ore with charcoal to produce sponge iron powders.
- These powders were subsequently formed into useful shapes by forging.
- Typical examples of ancient P/M artifacts include the **Ashoka Pillar of Delhi** and the even older Inca articles.
- Became evident as early as **19th century** when **tungsten powders were produced and strengthened with thoria for use in electric lamp filaments**.

- The development of **tungsten carbide tools** was another important milestone in the growth of modern P/M industry.
- **Production of porous bearings** and bushes by P/M further spurred its growth. Subsequently, ferrous and nonferrous parts, composites as well as dispersion-strengthened alloys made by P/M came into use.
- Apart from finished products, P/M is also employed in a number of useful applications in modern technology.
- **Metal powders in printing, catalysts in chemical industries and in food industry.**
- Several important innovations that have taken place in material synthesis, **Mechanical Alloying (MA), Rapid Solidification Process (RSP) for powder production, Cold Isostatic Pressing (CIP) and Hot Isostatic Pressing (HIP) for component fabrication.**

STEPS IN POWDER METALLURGY

(i) Powder production.
(ii) Compaction.
(iii) Sintering.
(iv) Secondary and other finishing operations.

Powder Production

- The raw material for P/M components is the powder.
- These powders are engineered materials in the sense they are manufactured to precise specifications to facilitate subsequent processing.
- The powders used in P/M can be pure elements, elemental blends or pre alloyed powders.
- The choice of starting material is influenced by the product type and to a lesser extent by the fabrication process to be used.
- Several production methods are available for making powders.
  - Most common method is the atomization, which produces nearly 80% of the total volume of powders.

- Reduction of compounds is another widely used technique used for the production of iron, copper as well as tungsten and molybdenum.
- Electrolysis is yet another important powder production method used widely for making copper, iron and silver powders.
- Powders having the desired size range, shape and other characteristics are chosen to obtain the properties required for the end use.
- Powders along with additives, are mixed thoroughly using mixers. Lubricants (0.5–2% by weight of the charge) may be added prior to mixing to facilitate easy ejection of the compact and to minimize the wear of compaction tool.
- Typical lubricants include waxes or metallic stearates, graphite, stearates.
- Powders are then tested for properties like size, flow, density and compressibility before being compacted.
- Powder characterization is very important since powder characteristics influence both the compaction behaviour as well as the properties of green and sintered parts.

Compaction

- Compaction of powder mixtures is generally carried out using dies machined to close tolerances.
- Equipment used for compaction includes mechanical or hydraulic presses.
- Die design is important and must ensure easy ejection of compact.
- The powder type and its characteristics influence the compaction pressure.
- The basic purpose of compaction is to produce a green compact with sufficient strength to withstand further handling operations.
- The pressed part, usually called the **"green compact,"** is then taken for sintering.
- Consolidation of powders may also be carried out at high temperature.
- **Hot extrusion, hot pressing and Hot Isostatic Pressing (HIP) are examples of this category.**
- **These methods are used for critical metallic and ceramic components requiring near theoretical density.**

Sintering

- Sintering of the green compact is carried out in a furnace under a controlled atmosphere to bond the particles metallurgically.
- Sintering is carried out at temperatures about 70% of the absolute melting point of the material – **Solid phase sintering**
- Bonding occurs by diffusion of atoms, giving integrity to the compact.
- **In other words, sintering serves to consolidate the mechanically-bonded powders into a coherent body having the desired service properties.**
- Shrinkage occurs during sintering resulting in densification of the part.
- Sintering is normally done under a protective atmosphere.

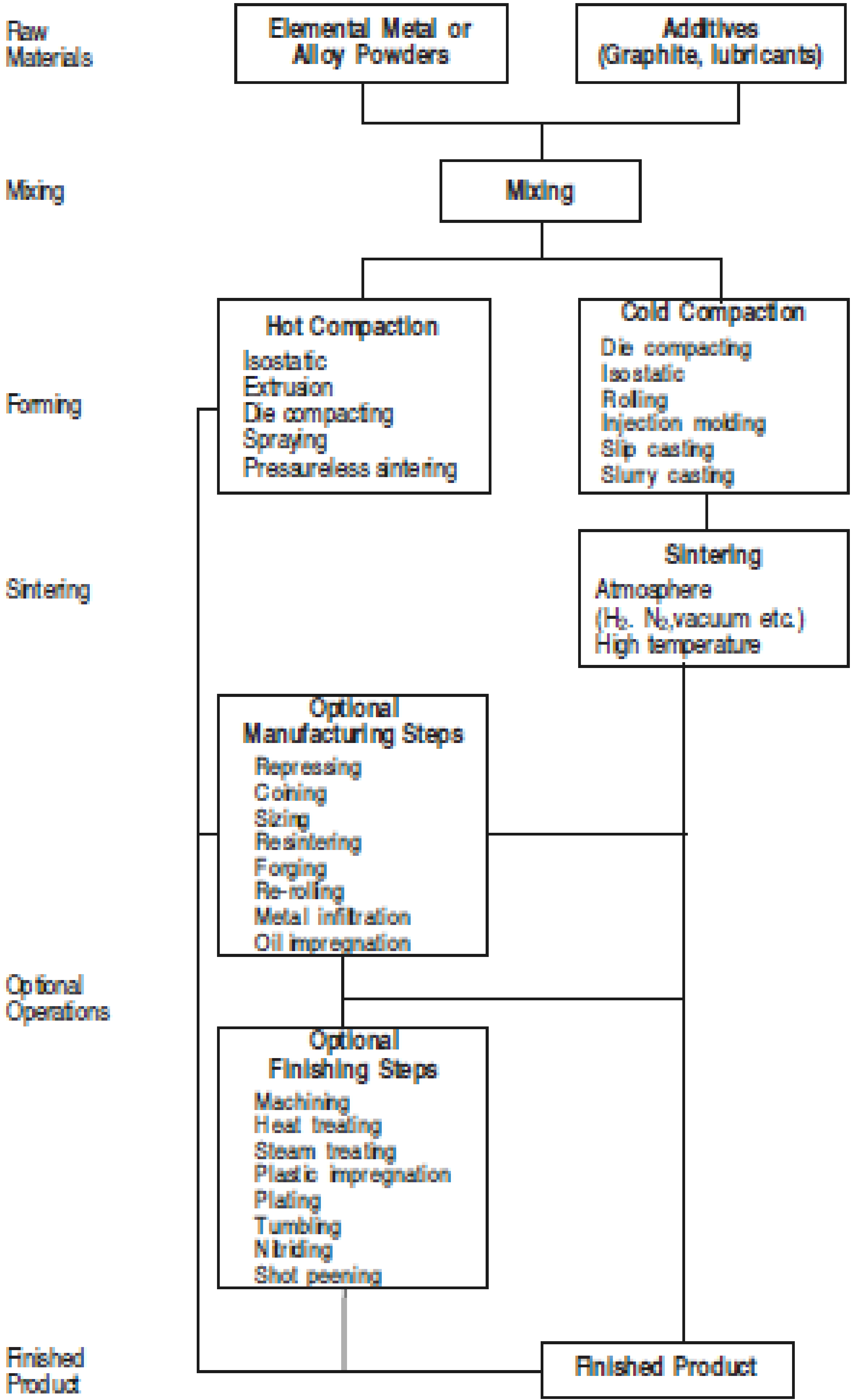
- **Liquid phase sintering,** involving the formation of a liquid phase to enhance sintering kinetics is another type of sintering.
- This type of sintering requires the presence of a low melting constituent in the compact, with heating being done above the melting point of this constituent.

- Other variations such as spark sintering, microwave sintering and laser sintering are also adopted for specific applications.
- Another important development is the Metal Injection Moulding (MIM), which has expanded the scope of P/M products tremendously

Secondary and Other Finishing Operations

- A sintered part may be subjected to one or more secondary or finishing operations.
- These are used to ensure close dimensional tolerances, improve surface finish, increase density and corrosion resistance.
- Typical operations include repressing, grinding and plating. Sometimes re-sintering is also done after repressing.
- Impregnation and infiltration are other operations used in the case of porous bearings and electrical contacts respectively.
- These products contain controlled amount of interconnected porosity, which are subsequently infiltrated with oil or other lower melting point metals by capillary action of the liquid phase.
- Oil impregnated bronze bearings and copper infiltrated tungsten.

- P/M parts can also be heat-treated.
- Steam treatment used for P/M parts is another method to improve the corrosion resistance as it effectively closes pores in iron and steel parts.
- Mechanical working of P/M parts after sintering may be done to improve dimensional tolerances and also for increasing the density.
- **P/M parts rarely need any machining, and hence it is essentially a 'chipless process', utilizing almost all the material.**



Raw Materials
Elemental Metal or Alloy Powders
Additives (Graphite, lubricants)
Mixing
Mixing
Forming
Hot Compaction
Isostatic
Extrusion
Die compacting
Spraying
Pressureless sintering
Cold Compaction
Die compacting
Isostatic
Rolling
Injection molding
Slip casting
Slurry casting
Sintering
Sintering
Atmosphere
(H_2, N_2,vacuum etc.)
High temperature
Optional Manufacturing Steps
Repressing
Coining
Sizing
Resintering
Forging
Re-rolling
Metal infiltration
Oil impregnation
Optional Operations
Optional Finishing Steps
Machining
Heat treating
Steam treating
Plastic impregnation
Plating
Tumbling
Nitriding
Shot peening
Finished Product
Finished Product

Process Advantages

Eliminates or minimises machining (little or no scrap) so efficient materials utilization—above 95% material utilization.
Enables close dimensional tolerances—near-net shapes possible.
Produces good surface finish.
Provides option for heat-treatment, for increasing strength or enhanced wear resistance and plating for improving corrosion resistance.
Facilitates manufacture of complex shapes, which would be impractical with other metal working processes.
Suited to moderate to high-volume component production requirements.
Components can be produced at reduced cost as compared to many other processes, i.e. cost-effective.
Components of hard materials which are difficult to machine can be readily manufactured, e.g. tungsten wires for incandescent lamps.
It is possible to produce components in pure form. Purity of the starting materials can be preserved throughout the process, a requirement for many critical applications.
Energy-efficient and Environment-friendly.

Metallurgical Advantages

Powders with uniform chemical composition with the desired characteristics, resulting from the absence of segregation during solidification. These characteristics will be reflected in the finished part.
Elemental and pre alloyed powders.
Unique compositions including **non-equilibrium compositions and** microstructures (crystalline, nanocrystalline and amorphous).
Wide variety of materials—metals and alloys of **miscible and immiscible** systems, refractory metals like tungsten and molybdenum, ceramics, polymers and composites.
Parts with controlled porosity.
Materials with improved magnetic property.

Limitations - Technical

Powder Making and Handling

Ratio of surface area to volume is very large and increases with increasing fineness - surface pollution either of chemical nature (such as oxidation) or of physical nature (such as adsorption) larger than for conventional parts
'clean environment,' is needed.

Compaction

Many compacting techniques such as cold or hot die pressing, cold or hot isostatic pressing, rolling, extrusion and forging have reached a high level of technical efficiency.
Die pressing allows a very good dimensional control of the finished part but imposes a residual porosity and density heterogeneities.

Isostatic pressing (with lesser shape limitations) and hot pressing can give theoretical densities but only at the expense of dimensional accuracy.
This technique is also costly and not applicable for all components.

Powder rolling, can be used only as a first step in a process.
Large thickness reduction is needed to reach close to theoretical density, powder rolling is limited to foil or thin sheet production.

Extrusion results in deep shear deformation and is favourable to attain high density and a perfect bonding. However, the original microstructure is replaced by a recrystallized fine grain with a fibrous texture.
Extrusion also can give only in a bar with a simple-shaped cross-section.

Powder forging brings a triaxial deformation of the blank at the expense of dimensional precision except in the case of isothermal forging, not available to many.

Sintering

- Sintering is made up of three stages, namely, the slow heating during burn-off, a constant temperature period during sintering and the cooling stage after sintering.
- The **non-steady and non-uniform temperature** conditions during heating and cooling can result in the formation of non-equilibrium structures and compositions.
- Such compositions may not attain equilibrium conditions during the constant temperature sintering phase.
- **Practically, a satisfactory control over the final microstructure is obtained more by empirical methods and not by any thorough theoretical procedures.**

Limitations – Economical

- In industrial practice, Powder Metallurgy is justified only when it allows a **cost-reduction**.
- In such a case, two different situations can arise:
- (i) the same **part can be made by an alternative route namely, casting, forging, machining** with similar characteristics in which case cost- comparison can be easily carried out. Depending on the cost-savings the procedure can be identified,
- (ii) in **cases where the same product cannot be produced by other techniques**, the cost-comparison is difficult to make.

- However, at present there are certain limitations in adopting Powder Metallurgy route due to part size and shape limitation dimensional precision, mechanical properties, alloy cleanliness and volume production.

Size Limitations

- Cost of tooling and of the presses required increase rapidly with size.
- Cost-efficiency in die pressing can be obtained only **for small parts weighing 20 to 100 g**ram and produced in large numbers.

- P/M, therefore, is **not recommended for large parts.**

- The other major limitation is on the **component design.**
- P/M components cannot accommodate certain features such as re-entrant angles or radial holes in vertically-pressed parts by fixed die pressing.

Mechanical Properties

- The presence of porosity introduces structural heterogeneity in die-pressed mechanical parts and hence the mechanical properties are inferior to wrought materials.
- Impurity content of powders is another factor which affects the properties of sintered parts.
- Vacuum or inert gas atomization and vacuum sintering can improve the mechanical properties.
- **However, in many cases the economics are against them**, unless the part is to be used in a critical application where cost is not a criterion.

- P/M components are **NOT recommended** for fatigue applications.

Volume Production

- Due to the cost of tooling for die pressing, its atomisation requires a sufficient number of parts to be made.
- For quantities **less than 10,000 parts, powder metallurgy has very little chance to be competitive against** other production techniques.
- The optimum number is considered to **be 1,00,000 parts.**

Limitations – Psychological

- In most cases, **Powder Metallurgy is a new production route** and hence if the parts currently being used are satisfactory, the user is **reluctant to change over to the Powder Metallurgy route**.

- Even when a new system is being designed, the **tendency to use 'classical materials'** most of the times by the designer precludes the use of **Powder Metallurgy material without even taking into account the specific possibilities** and advantages of Powder Metallurgy.

Many of the above limitations are likely to be overcome in the near future resulting in increased use of P/M parts in all spheres of application.